

## **Supplementary material**

### **EmbedSLR: Extended Bibliometric Validation Across Research Domains**

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This Supplementary Material complements the main article EmbedSLR: An Open-Source Python Framework for Efficient Embedding-Based Screening and Bibliometric Validation in Systematic Literature Review.

It presents an extended bibliometric validation of the EmbedSLR method conducted on five additional research problems from diverse scientific disciplines, beyond the illustrative example discussed in the main manuscript.

All selection procedures were carried out using the SBERT embedding backend with the all-distilroberta-v1 model, ensuring consistency with the methodology described in the main paper. The bibliometric indicators were computed for both the original Scopus datasets and for the subsets of 17 publications selected according to the lowest cosine distance from the natural-language problem descriptions.

The purpose of these supplementary analyses is to evaluate the generality, reproducibility, and robustness of the proposed approach in identifying thematically coherent subsets of publications in various contexts.

The document follows the same structure and metrics as used in the main paper. For each case, bibliometric indicators are calculated for two sets of articles:

1. the whole set of publications retrieved directly from Scopus based on keyword queries, and
2. a subset of 17 publications selected according to the lowest cosine distance using the embedding-based ranking.

These results demonstrate that EmbedSLR consistently improves the internal bibliometric coherence of publication subsets both in terms of shared references and shared keywords confirming its applicability across scientific domains.

The included charts (Figures 1-4) visualize average and Jaccard-based bibliometric indicators for all five cases, while detailed numerical results illustrate the magnitude of improvement after semantic selection.

## **DATA**

The method was tested on five cases representing distinct scientific fields and research problems. Each case is discussed in detail to illustrate the versatility of the proposed approach.

### **3.1. Case 1: Artificial intelligence in marketing**

**Research problem:** How is artificial intelligence (AI) used in companies' marketing activities? The increasing availability of AI tools enables the automation and personalization of marketing processes. However, it remains unclear to what extent AI applications genuinely improve marketing efficiency, decision-making, and customer engagement.

**Scopus query:** TITLE-ABS-KEY("artificial intelligence" AND "marketing")

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**Number of found records: 6761**

This dataset includes research focusing on the integration of AI into marketing strategy, analytics, and consumer interaction. It encompasses studies on recommendation systems, chatbots, and data-driven campaign optimization.

### **3.2. Case 2: Non-destructive testing (NDT) of reinforcement corrosion in reinforced concrete structures**

**Research problem:** How are non-destructive testing (NDT) methods used to assess reinforcement corrosion in reinforced concrete (RC) structures? Corrosion remains a critical durability issue in RC structures, and reliable NDT methods are essential for early detection and maintenance planning.

**Scopus query:** ALL(("reinforced concrete" OR "RC structure\*" OR (concrete W/2 (reinforc\* OR rebar\* OR "steel bar\*" OR "steel reinforcement"))) AND (corrosion OR "rebar corrosion" OR "steel corrosion" OR "corrosion-induced" OR debond\* OR delaminat\*) AND ("non-destructive" OR nondestructive OR "non destructive" OR NDT) AND ("eddy current" OR "pulsed eddy current" OR PEC OR "remote field eddy current" OR RFEC OR "magnetic flux leakage" OR MFL OR magnetometr\* OR "residual magnetization" OR "infrared thermograph\*" OR IRT OR "active thermograph\*" OR "microwave infrared thermography" OR MIRT OR "ground penetrating radar" OR GPR OR terahertz OR THz OR radiograph\* OR "X-ray" OR "X ray" OR gamma OR "magnetic force induced vibration" OR "M5 method"))

**Number of found records: 5245**

The publications collected cover various NDT techniques (e.g., GPR, thermography, eddy current) used to diagnose reinforcement corrosion. They provide a basis for examining the thematic consistency of research in structural diagnostics.

### **3.3. Case 3: Artificial intelligence in production automation**

**Research problem:** How is artificial intelligence used to automate production processes in industrial settings? AI-driven automation is transforming modern manufacturing, but the literature is fragmented across technological and managerial perspectives.

**Scopus query:** TITLE-ABS-KEY("automation" AND "production" AND "artificial intelligence")

**Number of found records: 3087**

This case investigates AI applications in production lines, predictive maintenance, and process optimization, exploring both the technological potential and organizational challenges of industrial automation.

### 3.4. Case 4: The impact of the ketogenic diet on health

**Research problem:** What is the impact of the ketogenic diet on human health? While the ketogenic diet has gained popularity for its metabolic and therapeutic benefits, its long-term effects and safety remain debated.

**Scopus query:** TITLE-ABS-KEY("ketogenic diet" AND "health")

**Number of found records:** 1765

This dataset comprises studies from medical and nutritional sciences analyzing physiological effects, potential risks, and clinical outcomes associated with ketogenic diets.

### 3.5. Case 5: The impact of artificial intelligence on young people's mental health

**Research problem:** What is the impact of artificial intelligence on the mental health of young people? AI applications, particularly in social media and digital content delivery, may affect emotional well-being, social interaction, and psychological development.

**Scopus query:** TITLE-ABS-KEY("impact" AND "AI" AND "young" AND "health")

**Number of found records:** 271

This collection includes interdisciplinary research addressing ethical, psychological, and social aspects of AI exposure among youth populations, particularly concerning mental health risks and digital well-being.

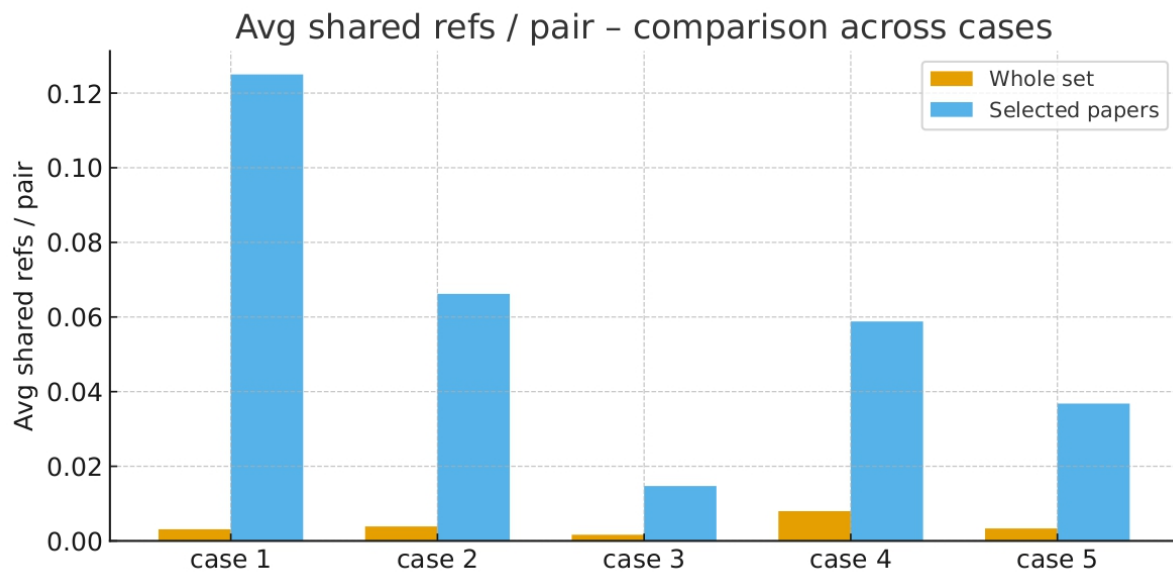
## RESULTS

The bibliometric analysis was conducted for five independent cases to evaluate the effectiveness of the proposed method in identifying thematically consistent subsets of publications. For each case, two sets of articles were analyzed:

- (1) the whole set returned from Scopus in response to the query and
- (2) a subset of 17 publications selected based on semantic similarity to the problem description (lowest cosine distance).

The comparison includes four indicators: *average number of shared references per pair (AR)*, *Jaccard index for references (JR)*, *average number of shared keywords per pair (AK)*, and *Jaccard index for keywords (JK)*.

Figure 1. Average number of shared references per pair (AR)



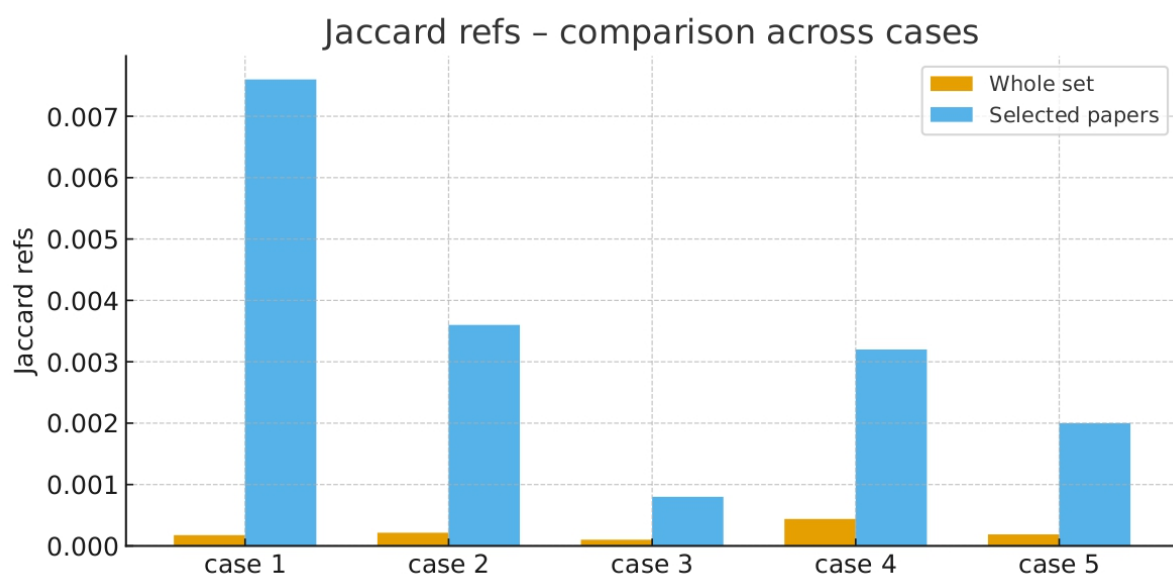
Source: own elaboration.

The *Avg shared refs / pair* values in fig.1 increased markedly for all cases after the selection procedure.

- **Case 1 (AI in marketing):** from 0.0031 to 0.125 (+3,900%)
- **Case 2 (NDT in RC corrosion):** from 0.0039 to 0.0662 (+1,606%)
- **Case 3 (AI in production):** from 0.0017 to 0.0147 (+789%)
- **Case 4 (Ketogenic diet & health):** from 0.0080 to 0.0588 (+638%)
- **Case 5 (AI & youth mental health):** from 0.0033 to 0.0368 (+1,003%)

Such increases confirm that the selected subsets exhibit a substantially higher density of shared citations, indicating stronger bibliographic connectivity and thematic coherence within the selected publications.

Figure 2. Jaccard Index for references (JR)



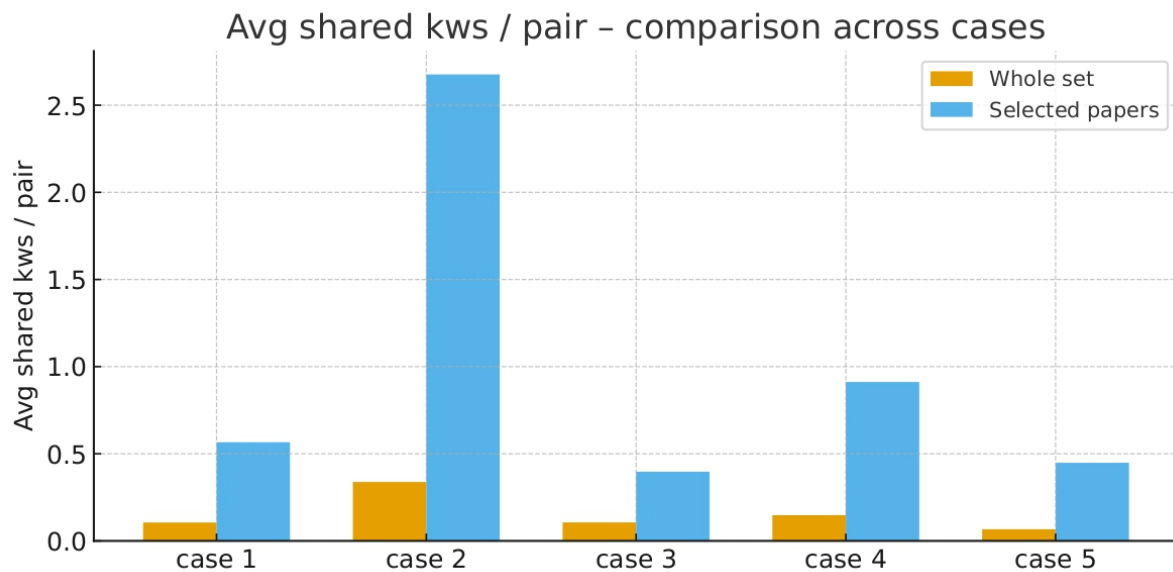
Source: own elaboration.

The *Jaccard refs* indicator in fig. 1 also rose for all cases:

- **Case 1:** from 0.00018 to 0.0076 (+4,238%),
- **Case 2:** from 0.00021 to 0.0036 (+1,578%),
- **Case 3:** from 0.00010 to 0.0008 (+692%),
- **Case 4:** from 0.00044 to 0.0032 (+629%),
- **Case 5:** from 0.00019 to 0.0020 (+965%).

Although absolute values are small as expected for pairwise reference overlaps the percentage increases clearly demonstrate stronger bibliographic similarity among the selected papers.

Figure 3. Average number of shared keywords per pair (AK)



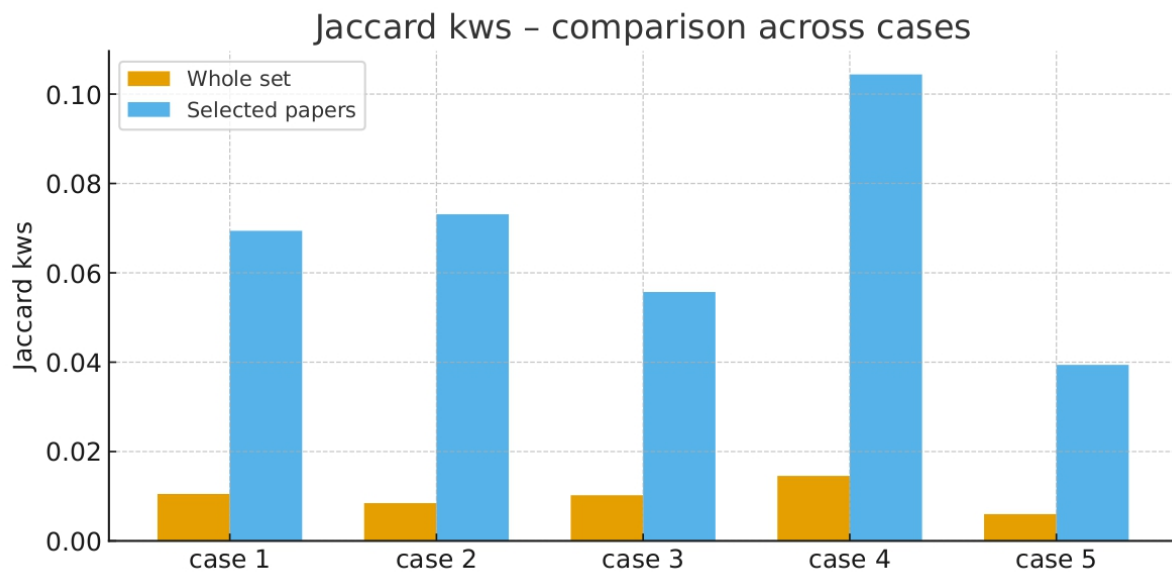
Source: own elaboration.

The keyword-based indicators in fig. 3 show even more substantial improvements:

- **Case 1:** from 0.106 to 0.566 (+434%),
- **Case 2:** from 0.338 to 2.677 (+692%),
- **Case 3:** from 0.106 to 0.397 (+274%),
- **Case 4:** from 0.148 to 0.912 (+518%),
- **Case 5:** from 0.067 to 0.449 (+572%).

These large increases suggest that the selected subsets are not only bibliographically closer but also semantically concentrated around shared terminology, demonstrating the high efficiency of the semantic selection process.

Figure 4. Jaccard Index for keywords (JK)



Source: own elaboration.

The *Jaccard kws* values in fig. 4 improved in all cases:

- **Case 1:** from 0.0105 to 0.0694 (+561%),
- **Case 2:** from 0.0085 to 0.0731 (+764%),
- **Case 3:** from 0.0102 to 0.0557 (+445%),
- **Case 4:** from 0.0145 to 0.1044 (+619%),
- **Case 5:** from 0.0060 to 0.0394 (+557%).

These results confirm that the subset selection based on embeddings significantly improves terminological overlap and topic alignment.

Across all five cases, each bibliometric indicator increased substantially after applying the embedding-based selection procedure.

- The **average shared references** and **Jaccard reference indices** improved by factors ranging from +600% to +4,000%.
- The **keyword-based indicators** showed slightly smaller but still strong gains of +270% to +760%.

These results provide a comprehensive validation of the EmbedSLR software, demonstrating its robustness and effectiveness across diverse scientific domains. The consistent improvements observed in all tested cases indicate that the method not only filters and concentrates thematically related articles but also ensures higher semantic integrity and bibliometric coherence. The findings confirm that EmbedSLR constitutes a reliable and generalizable framework for conducting systematic literature reviews with enhanced accuracy and reproducibility.

Link to GitHub where all files used in this study can be found: <https://github.com/s-matysik/EmbedSLR/tree/main/additional-validation>